UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Artificial Intelligence

TOPIC 12:
SUPERVISED LEARNING MODELLING AND EVALUATION: CLASSIFICATION USING ARTIFICIAL NEURAL NETWORK

MACHINE LEARNING (SAIA 2113)

Session 2025/2026 - Sem 1

Prepared by:
Dr. Noor Jannah Zakaria
noor.jannah@utm.my

Innovating Solutions

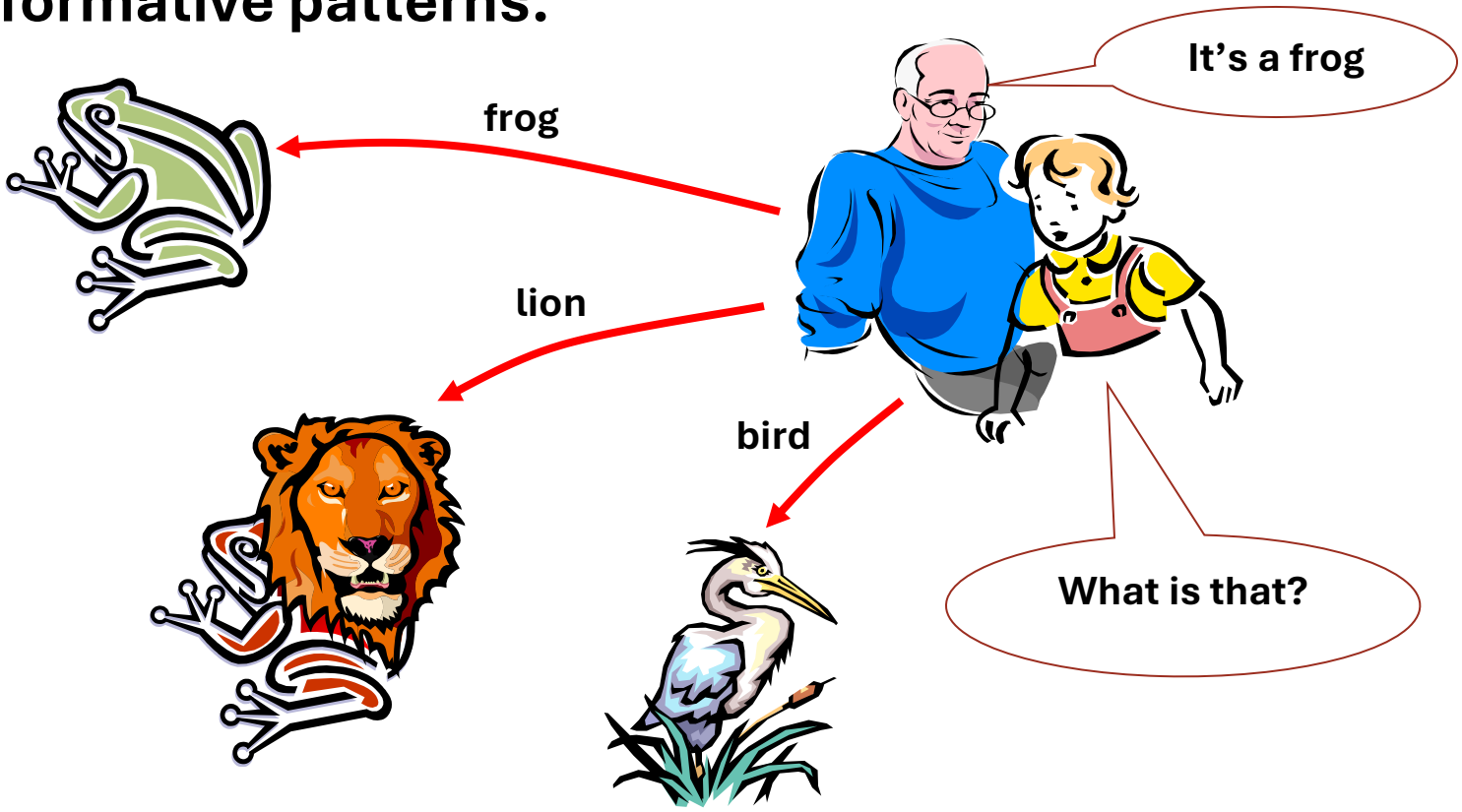
Outline

- What is Artificial Neural Network ?
- Biological Neuron
- Comparing ANN & BNN
- Model
- Architecture
- Learning
- Advantages & Disadvantages
- Applications
- Mathematical Representation
- Backpropagation & Stopping Criteria
- ANN Design Issues

www.fai.utm.my

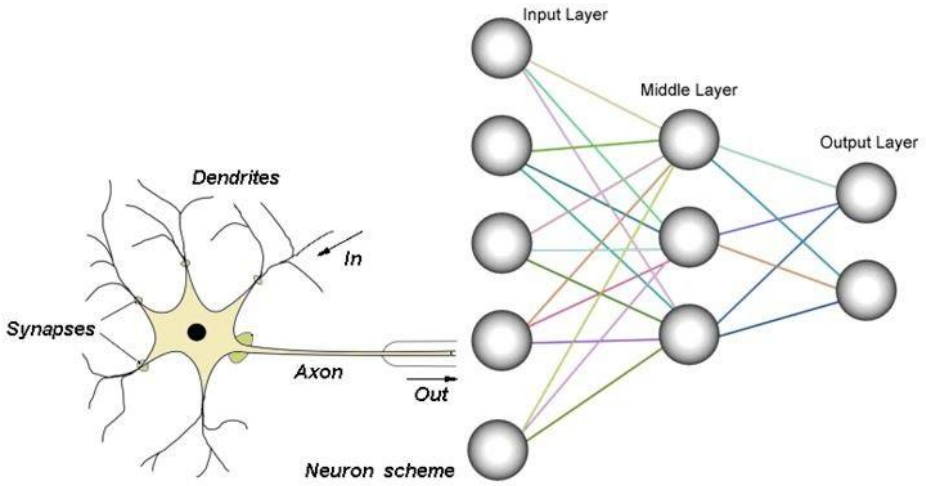
The idea of ANNs?

- NNs learn relationship between cause and effect or organize large volumes of data into orderly and informative patterns.



WHAT IS ANN?

- Data processing system consisting of a large number of simple, highly interconnected processing elements (artificial neurons) in an architecture inspired by the structure of the cerebral cortex of the brain.



What is Artificial Neural Network ?

- Computing system whose central theme is borrowed from the analogy of biological neural networks.
- Named as “artificial neural systems,” or “parallel distributed processing systems,” or “connectionist systems.”
- Acquires a large collection of units that are interconnected in some pattern to allow communication between the units.
- These units, also referred to as nodes or neurons, are simple processors which operate in parallel.

What is Artificial Neural Network ?

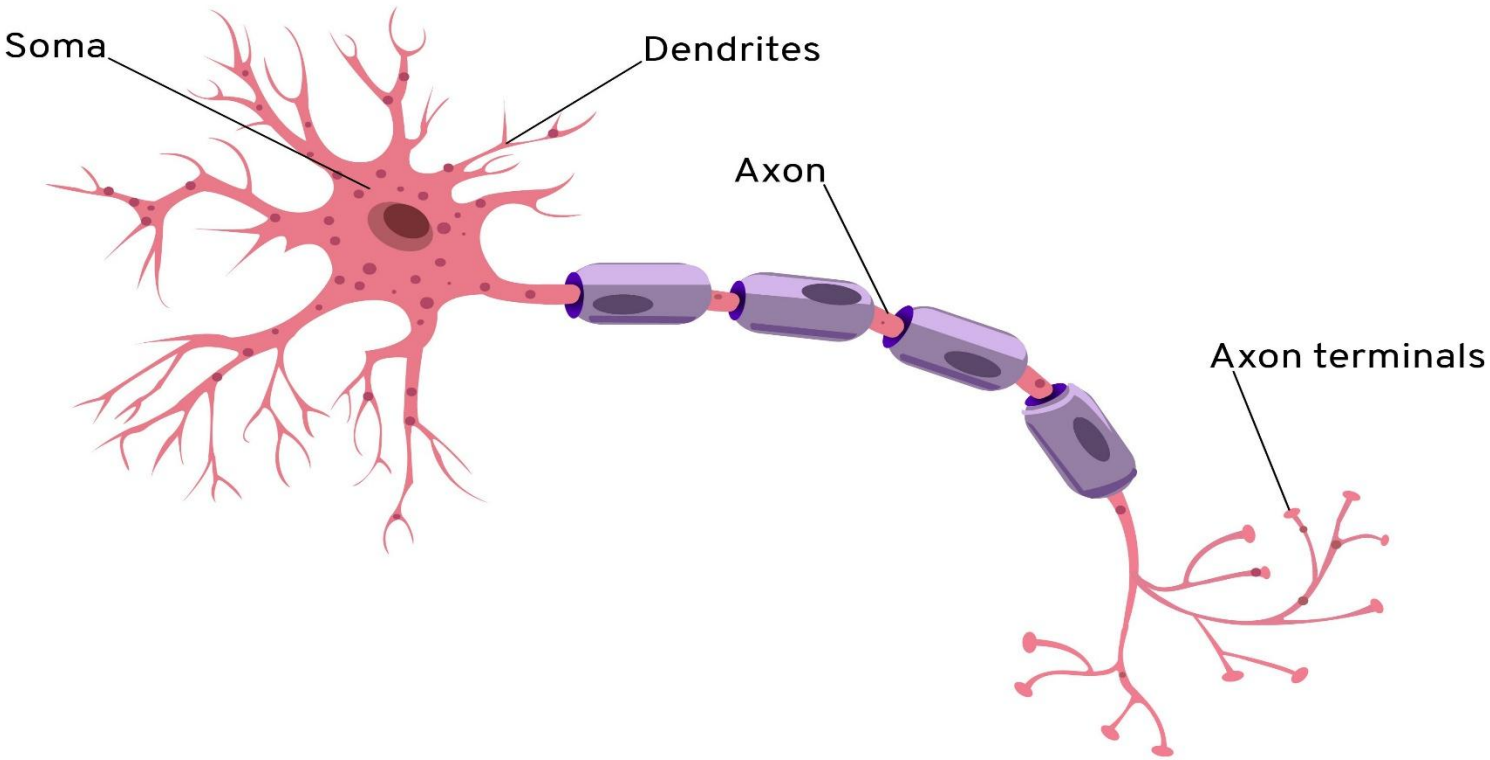
- Every neuron is connected with other neuron through a connection link.
- Each connection link is associated with a weight that has information about the input signal.
- This is the most useful information for neurons to solve a particular problem because the weight usually excites or inhibits the signal that is being communicated.
- Each neuron has an internal state, which is called an activation signal.
- Output signals, which are produced after combining the input signals and activation rule, may be sent to other units.

Outline

- What is Artificial Neural Network ?
- Biological Neuron
- Comparing ANN & BNN
- Model
- Architecture
- Learning
- Advantages & Disadvantages
- Applications
- Mathematical Representation
- Backpropagation & Stopping Criteria
- ANN Design Issues

www.fai.utm.my

Human Biological Neuron



Biological Neuron

- A neuron (also known as a neurone or nerve cell): Cell that carries electrical impulses.

- Neurons are the basic units of the nervous system and its most important part is the brain.

- Every neuron is made of a cell body (also called a soma), dendrites and an axon.

Biological Neuron

- Dendrites and axons are nerve fibres. There are about 86 billion neurons in the human brain, which comprises roughly 10% of all brain cells. The neurons are supported by glial cells and astrocytes.

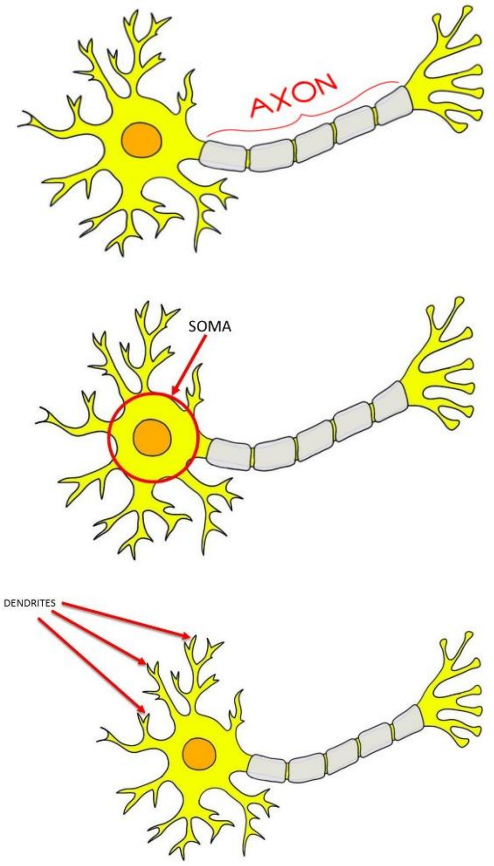
- Neurons are connected to one another and tissues. They do not touch and instead form tiny gaps called synapses.

- These gaps can be chemical synapses or electrical synapses and pass the signal from one neuron to the next.

Biological Neural Networks

- **A Biological neuron has three components :**

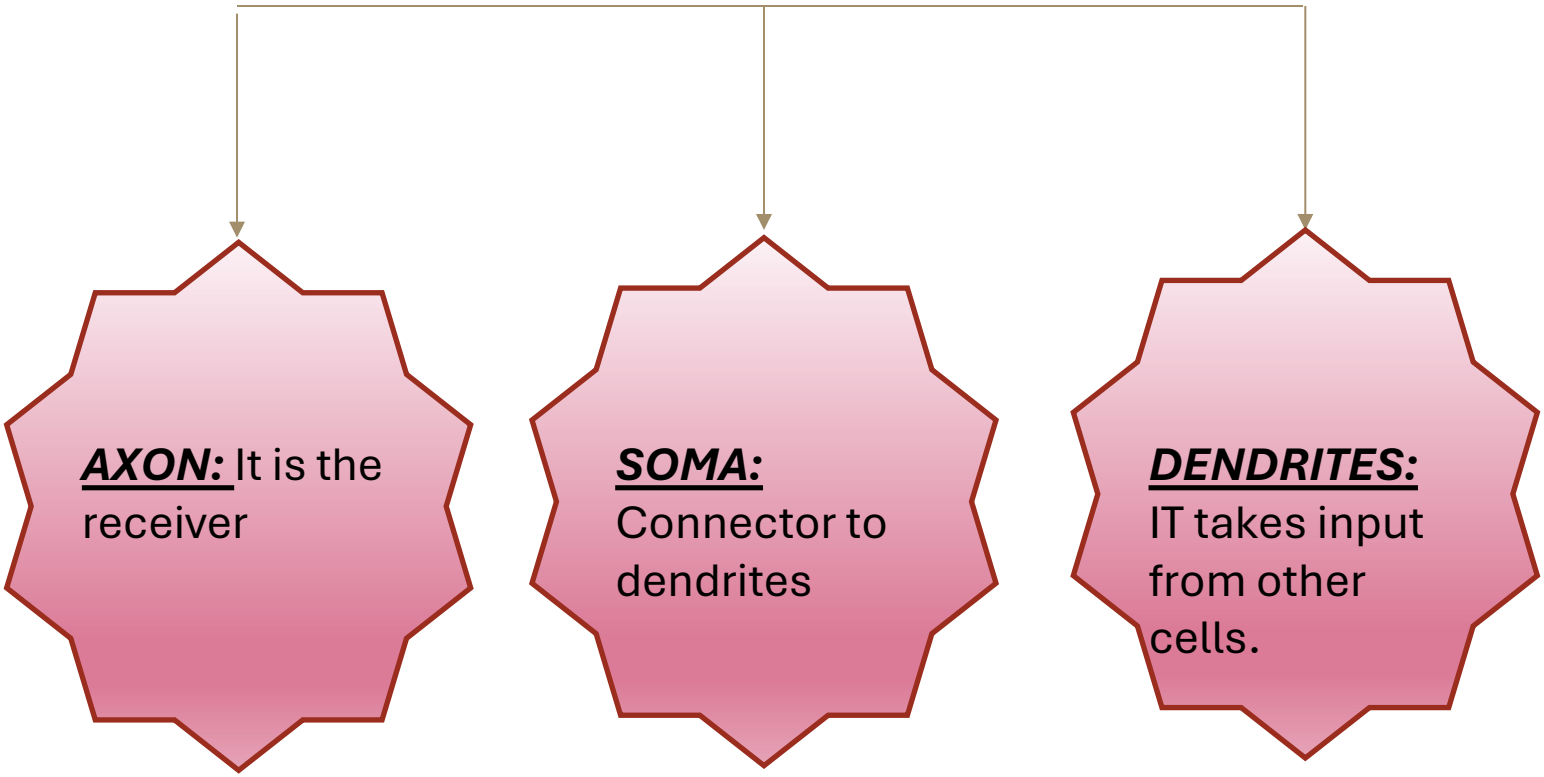
- AXON
- SOMA(CELL BODY)
- DENDRITES



Biological Neuron

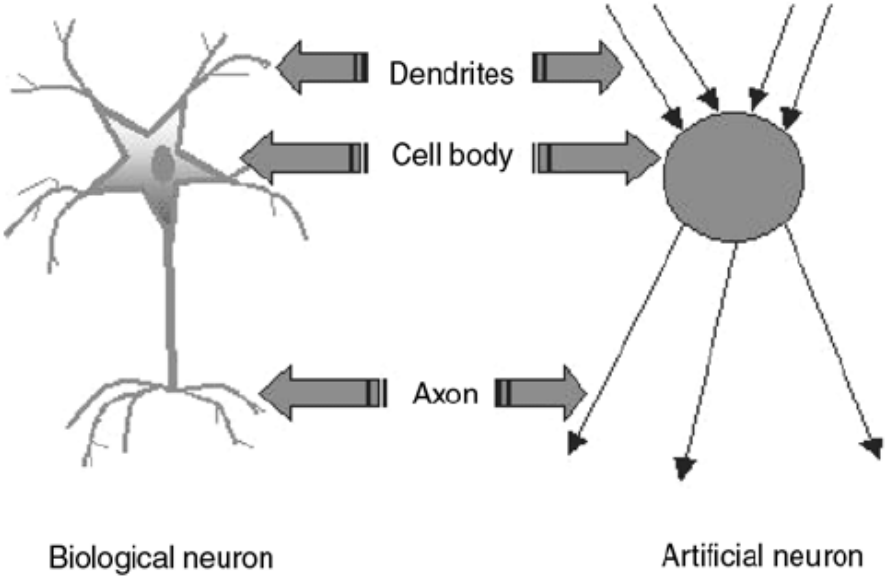
- Dendrite—Receives signals from other neurons.
- Soma (cell body)—Sums all the incoming signals to generate input.
- Axon—When the sum reaches a threshold value, neuron fires and the signal travels down the axon to the other neurons.
- Synapses—The point of interconnection of one neuron with other neurons. The amount of signal transmitted depend upon the strength (synaptic weights) of the connections.
- The connections can be inhibitory (decreasing strength) or excitatory (increasing strength) in nature.
- So, neural network, in general, is a highly interconnected network of billions of neuron with trillion of interconnections between them.

WHAT AM I MADE OF?



A Graphical Analogy

- A physical neuron
- An artificial neuron



Are we the same?

ANN	***Biological Neuron***
INPUT	DENDRITES
NODE	SOMA
OUTPUT	AXON
INTERCONNECTION	SYNAPSE

Outline

- What is Artificial Neural Network ?
- Biological Neuron
- Comparing ANN & BNN
- Model
- Architecture
- Learning
- Advantages & Disadvantages
- Applications
- Mathematical Representation
- Backpropagation & Stopping Criteria
- ANN Design Issues

www.fai.utm.my

How is Brain Different from Computers

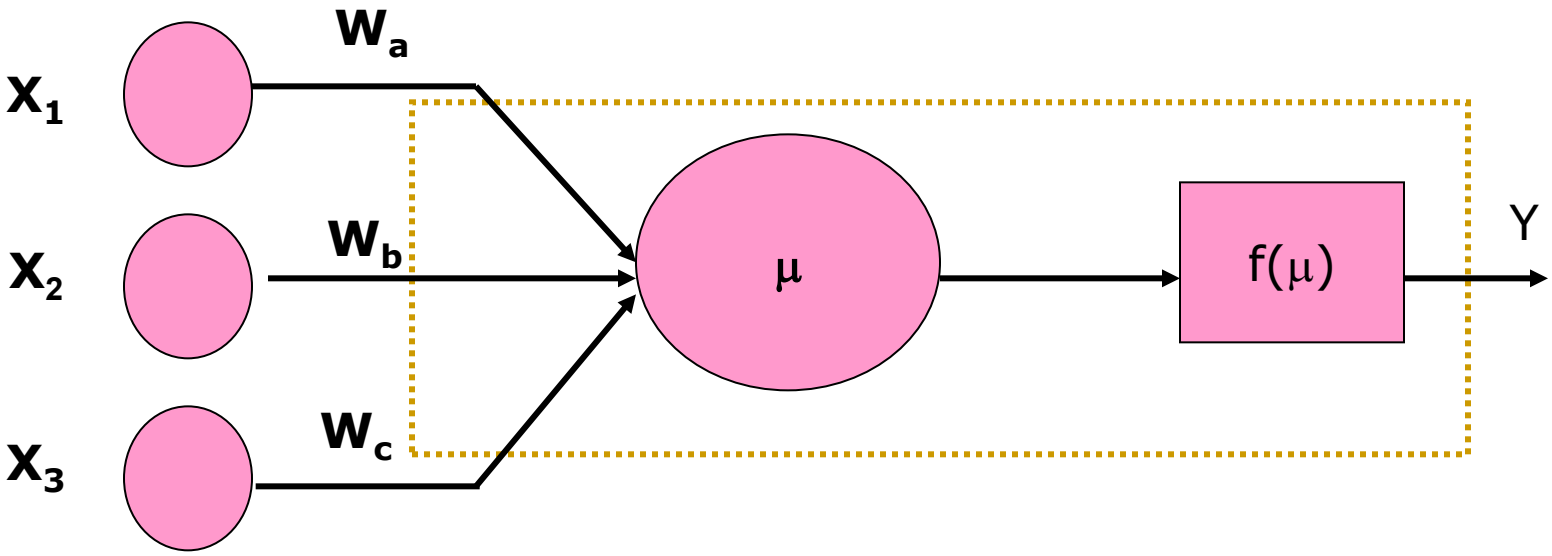
BRAIN	COMPUTERS
Biological Neurons or Nerve Cells.	Silicon Transistor.
200 Billion Neurons, 32 trillion interconnections.	1 Billion bytes RAM, trillion of bytes on disk.
Neuron Size: 10-6m.	Single Transistor Size: 10-9m.
Energy Consumption: 6-10 Joules operation per second.	Energy Consumption: 10-16 Joules Operation per second
Learning Capability	Programming capability

Comparing ANN with BNN

- As this concept borrowed from ANN there are lot of similarities though there are differences too.
- Similarities are in the following table

Biological Neural Network	Artificial Neural Network
Soma	Node
Dendrites	Input
Synapse	Weights or Interconnections
Axon	Output

Model Of A perceptron



Input units

Connection weights

Summing function

computation

(dendrite)

(synapse)

(soma)

(axon)

Comparing ANN with BNN

Following table Shows some differences between ANN and BNN :-

Criteria	BNN	ANN
Processing	Massively parallel, slow but superior than ANN	Massively parallel, fast but inferior than BNN
Size	1011 neurons and 1015 interconnections	102 to 104 nodes (mainly depends on the type of application and network designer)
Learning	They can tolerate ambiguity	Very precise, structured and formatted data is required to tolerate ambiguity
Fault tolerance	Performance degrades with even partial damage	It is capable of robust performance, hence has the potential to be fault tolerant
Storage capacity	Stores the information in the synapse	Stores the information in continuous memory locations

Analogy of ANN with BNN

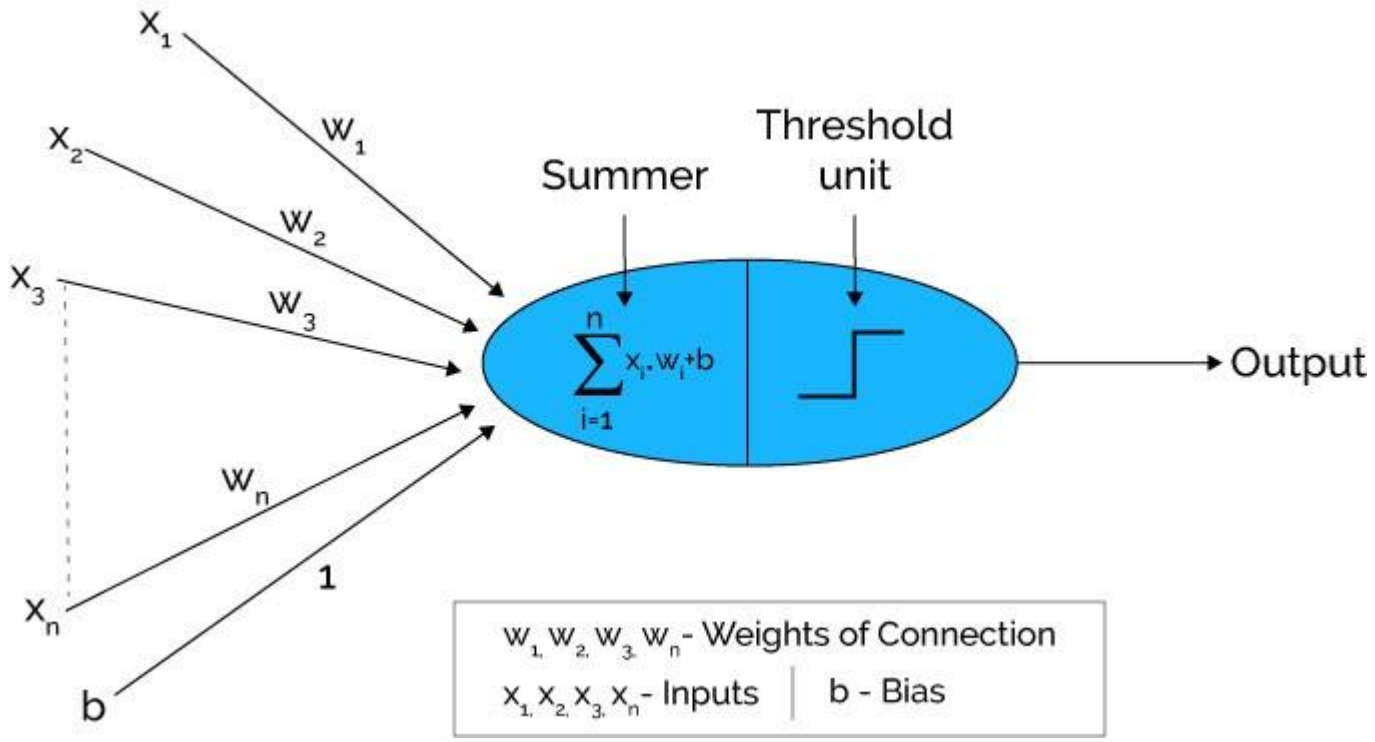
- The dendrites in biological neural network is analogous to the weighted inputs based on their synaptic interconnection in artificial neural network.
- Cell body is analogous to the artificial neuron unit in artificial neural network which also comprises of summation and threshold unit.
- Axon carry output that is analogous to the output unit in case of artificial neural network. So, ANN are modelled using the working of basic biological neurons.

Outline

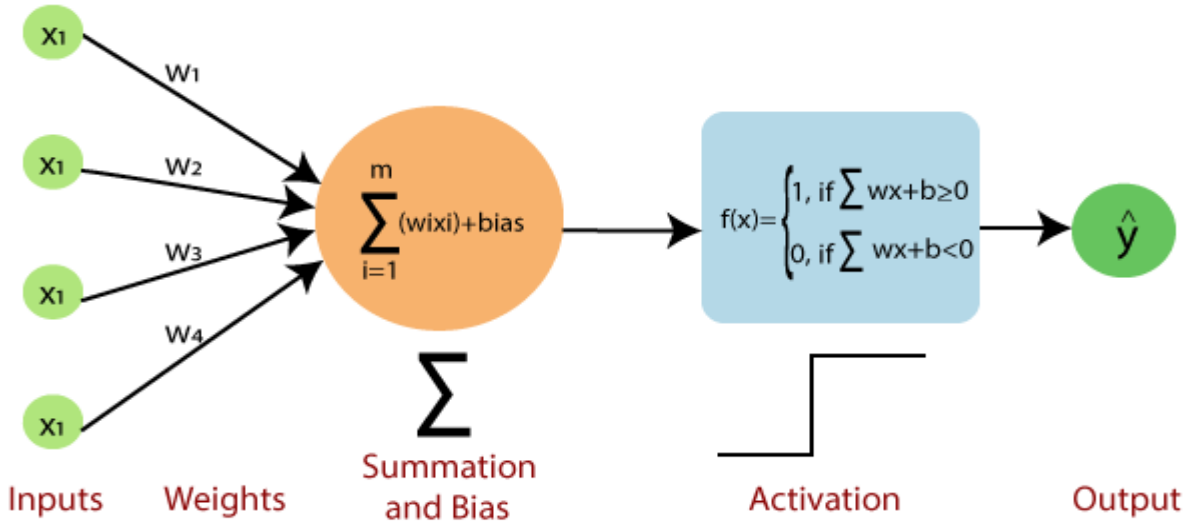
- What is Artificial Neural Network ?
- Biological Neuron
- Comparing ANN & BNN
- Model
- Architecture
- Learning
- Advantages & Disadvantages
- Applications
- Mathematical Representation
- Backpropagation & Stopping Criteria
- ANN Design Issues

www.fai.utm.my

Model of Artificial Neural Network



Artificial neuron is the basic unit of the neural network. In simple terms, it is a mathematical function based on a model of biological neurons. It can also be seen as a simple logic gate with binary outputs. They are sometimes also called ***perceptrons*.**



Model of Artificial Neural Network

- ANNs can be viewed as weighted directed graphs in which artificial neurons are nodes and directed edges with weights are connections between neuron outputs and neuron inputs.

- Receives input from the external world in the form of pattern and image in vector form.

- These inputs are mathematically designated by the notation x(n) for n number of inputs.

- Each input is multiplied by its corresponding weights.

Model of Artificial Neural Network

- Weights are the information used by the neural network to solve a problem.
- Typically weight represents the strength of the interconnection between neurons inside the neural network.
- The weighted inputs are all summed up inside computing unit (artificial neuron).
- In case the weighted sum is zero, bias is added to make the output not zero or to scale up the system response.
- Bias has the weight and input always equal to ‘1’.

Model of Artificial Neural Network

- The sum corresponds to any numerical value ranging from 0 to infinity.

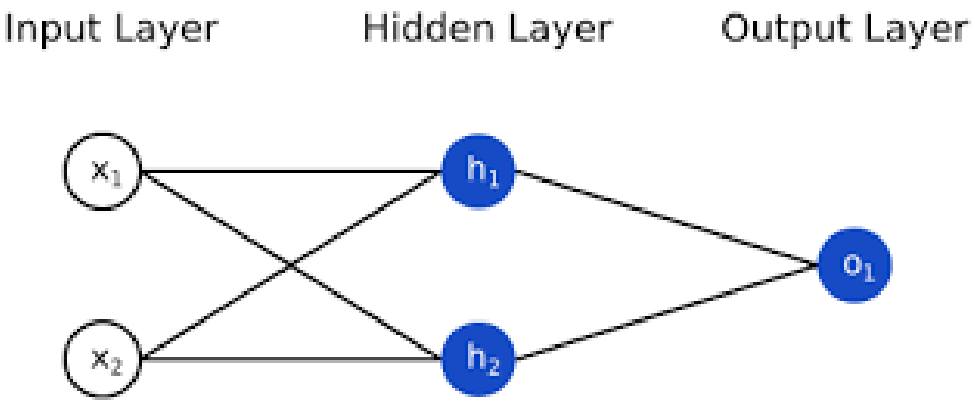
- In order to limit the response to arrive at desired value, the threshold value is set up. For this, the sum is passed through activation function.

- The activation function is set of the transfer function used to get desired output. There are linear as well as the non-linear activation function.

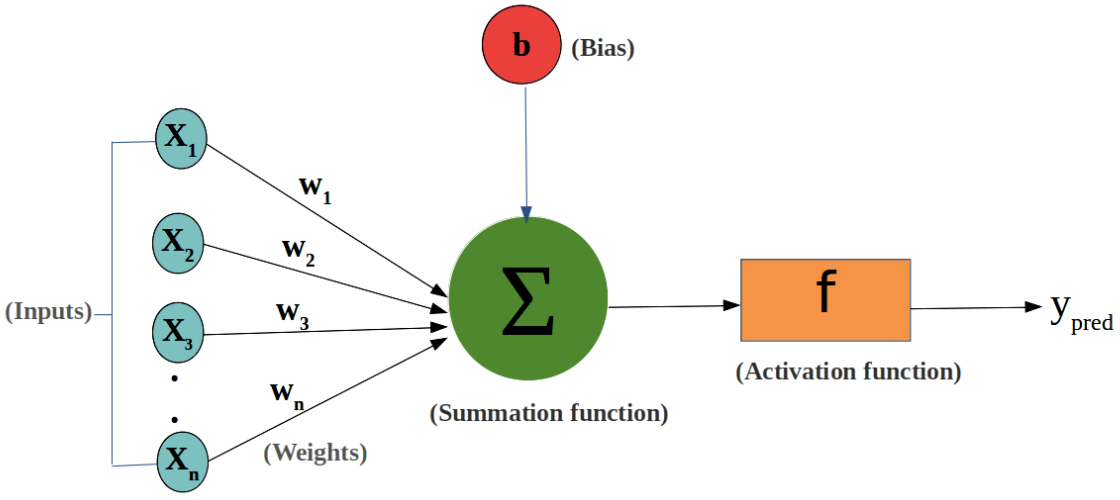
Model of Artificial Neural Network

- Some of the commonly used activation function are—binary, sigmoidal (linear) and tan hyperbolic sigmoidal functions(nonlinear).
- Binary—The output has only two values either 0 and 1. For this, the threshold value is set up. If the net weighted input is greater than 1, an output is assumed 1 otherwise zero.
- Sigmoidal Hyperbolic—This function has ‘S’ shaped curve. Here tan hyperbolic function is used to approximate output from net input. The function is defined as—f (x) = (1/1+ exp(-σx)) where σ—steepness parameter.

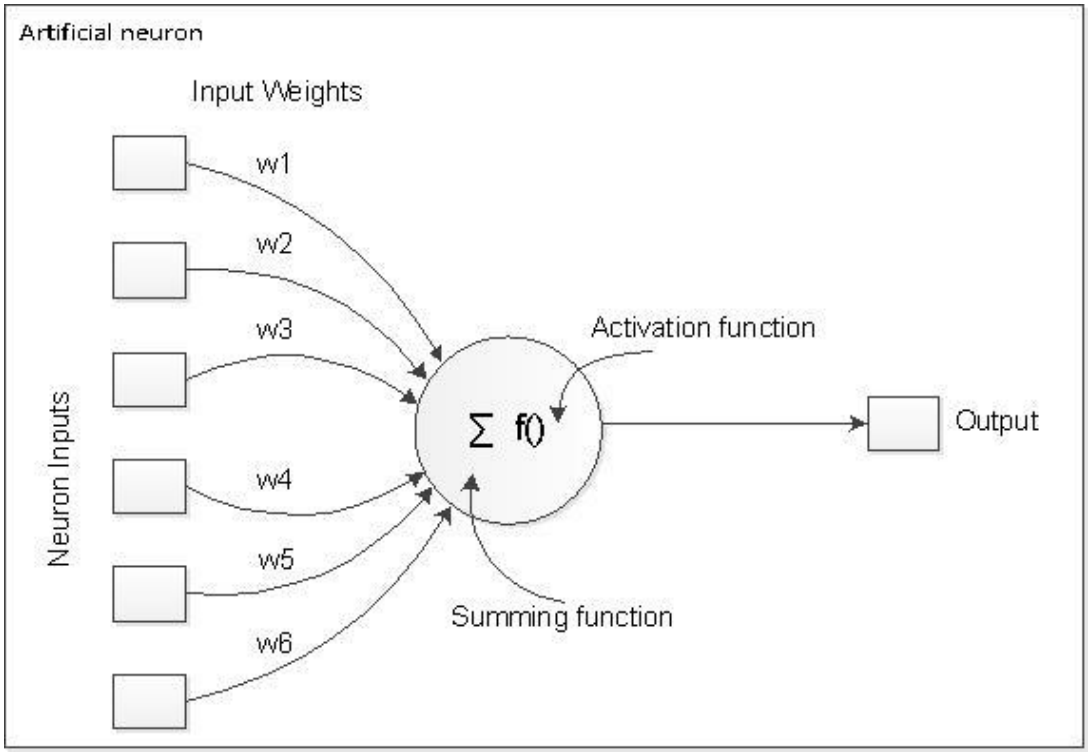
- ***INPUT VALUES*** **>>**We pass input values to a neuron using this layer. It might be something as simple as a collection of array values. It is similar to a dendrite in biological neurons .



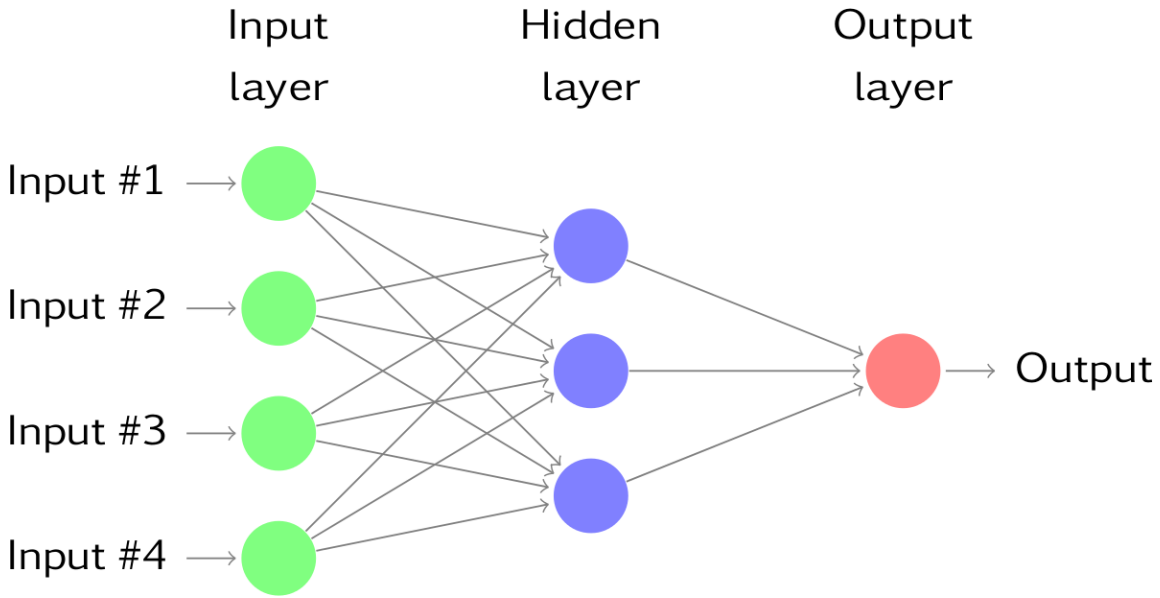
- ***WEIGHT AND BIAS >>*** Weights are a collection of array values which are multiplied to the respective input values. We then take a sum of all these multiplied values which is called a weighted sum. Next, we add a bias value to the weighted sum to get final value for prediction by our neuron.



- ***ACTIVATION FUNCTION >>*** Activation Function decides whether or not a neuron is fired. It decides which of the two output values should be generated by the neuron.



- ***OUTPUT LAYER >>*** Output layer gives the final output of a neuron which can then be passed to other neurons in the network or taken as the final output value.



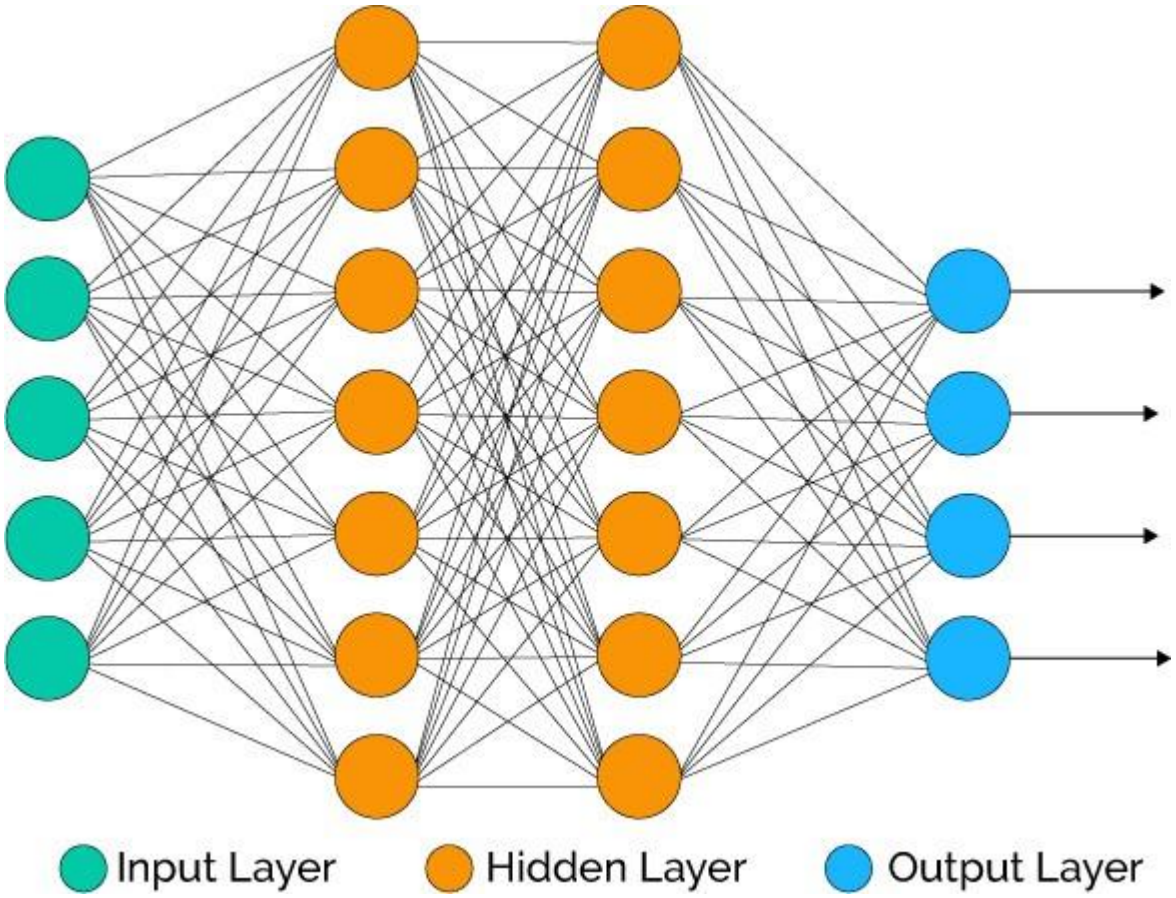
Outline

- What is Artificial Neural Network ?
- Biological Neuron
- Comparing ANN & BNN
- Model
- Architecture
- Learning
- Advantages & Disadvantages
- Applications
- Mathematical Representation
- Backpropagation & Stopping Criteria
- ANN Design Issues

www.fai.utm.my

Architecture

A typical neural network contains a large number of artificial neurons called units arranged in a series of layers. In typical artificial neural network, comprises different layers -



Architecture

- Input layer—It contains those units (artificial neurons) which receive input from the outside world on which network will learn, recognize about or otherwise process.

- Output layer—It contains units that respond to the information about how it's learned any task.

- Hidden layer—These units are in between input and output layers. The job of hidden layer is to transform the input into something that output unit can use in some way.

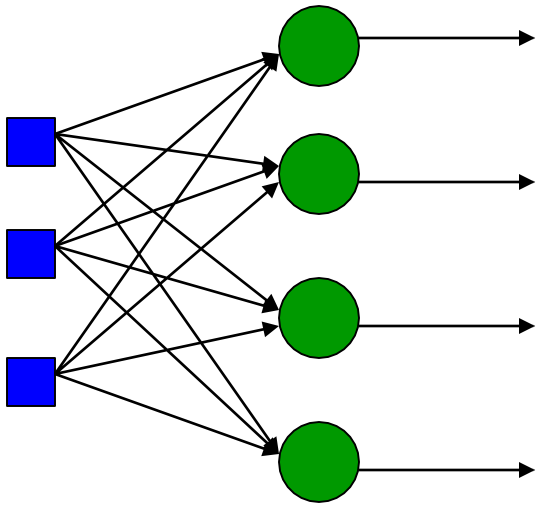
- Most neural networks are fully connected that means to say each hidden neuron is fully connected to the every neuron in its previous layer(input) and to the next layer (output) layer

Network Architectures

- Three different classes of network architectures
  - single-layer feed-forward
  - multi-layer feed-forward
  - recurrent
- The architecture of a neural network is linked with the learning algorithm used to train

Single Layer Feed-forward

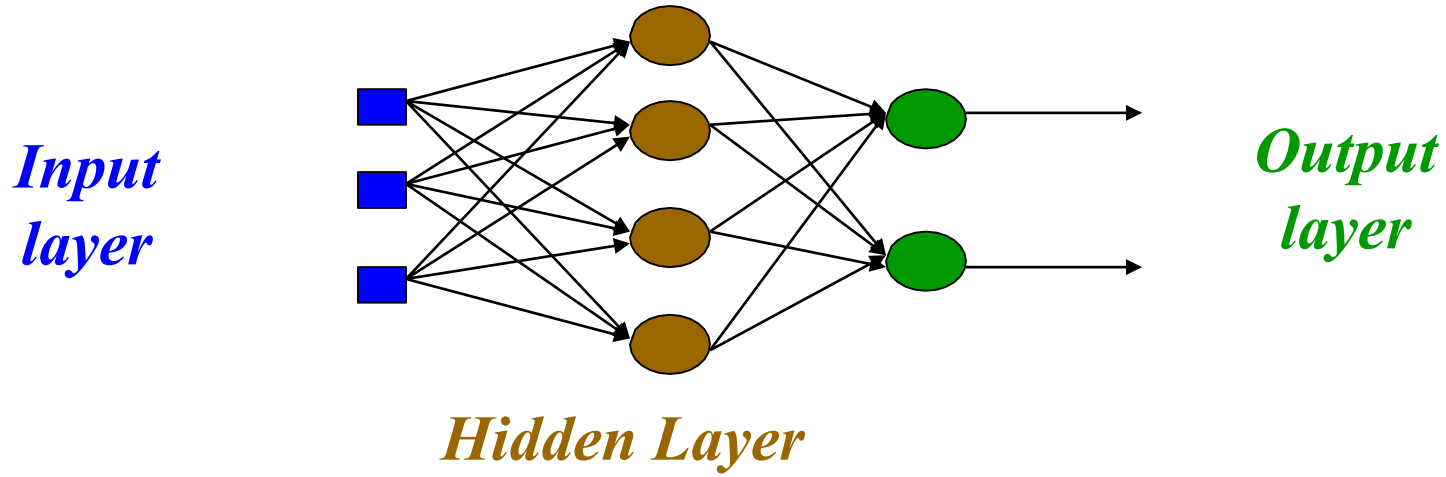
Input layer of source nodes



Output layer of neurons

Multi layer feed-forward NN (FFNN)

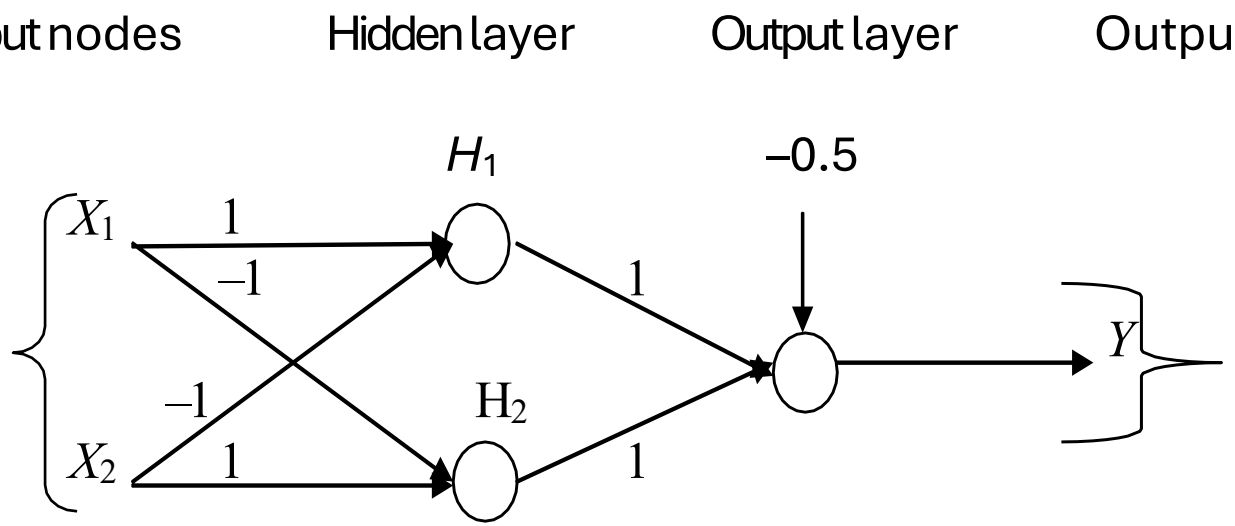
- FFNN is a more general network architecture, where there are hidden layers between input and output layers.
- Hidden nodes do not directly receive inputs nor send outputs to the external environment.
- FFNNs overcome the limitation of single-layer NN.
- They can handle non-linearly separable learning tasks.



3-4-2 Network

FFNN for XOR

- The ANN for XOR has two hidden nodes that realizes this non-linear separation and uses the sign (step) activation function.
- Arrows from input nodes to two hidden nodes indicate the directions of the weight vectors (1,-1) and (-1,1).
- The output node is used to combine the outputs of the two hidden nodes.



HOW DOES ANN WORKS

- Artificial neural networks (ANNs) are comprised of a node layers, containing an input layer, one or more hidden layers, and an output layer.
- Each node, or artificial neuron, connects to another and has an associated weight and threshold.
- Once an input layer is determined, weights are assigned. These weights help determine the importance of any given variable, with larger ones contributing more significantly to the output compared to other inputs.
- Their respective weights and then summed. Afterward, the output is passed through an activation function, which determines the output
- If the output of any individual node is above the specified threshold value, that node is activated, sending data to the next layer of the network.

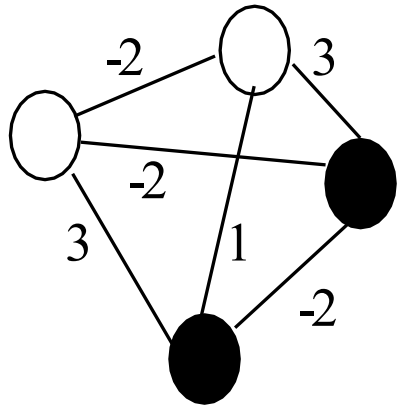
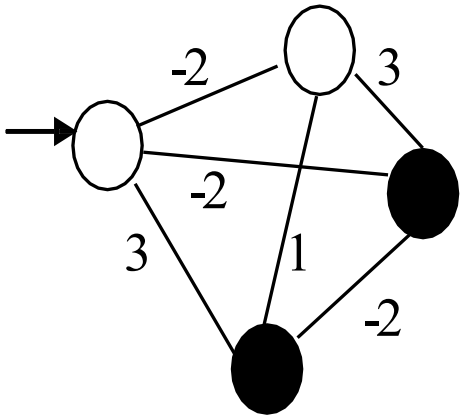
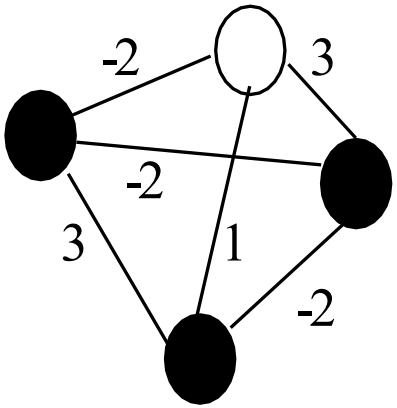
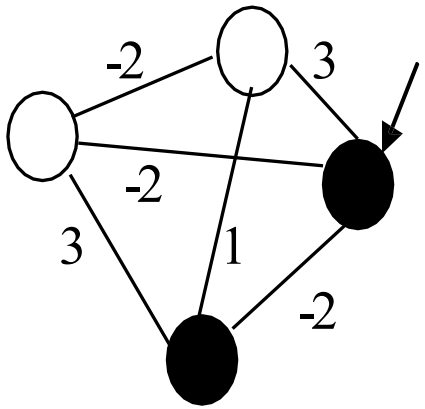
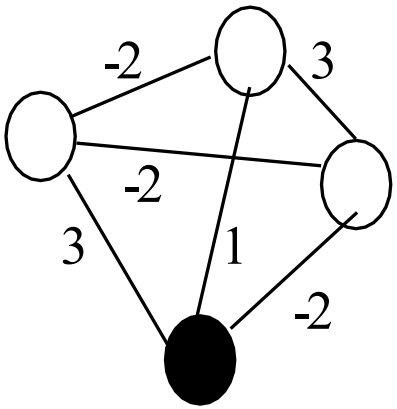
How Does ANN Learns?

- Typically, an ANN is initially trained or fed large amounts of data. Training consists of providing input and telling the network what the output should be.
- For example, to build a network that identifies the faces of actors, the initial training might be a series of pictures, including actors, non-actors, masks, statuary and animal faces.
- Each input is accompanied by the matching identification, such as actors' names or "not actor" or "not human" information. Providing the answers allows the model to adjust its internal weightings to learn how to do its job better.

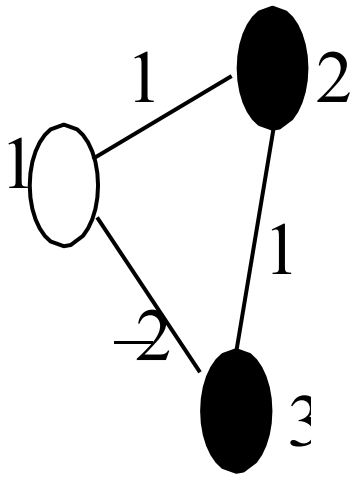
Activation Algorithm

Active unit represented by 1 and inactive by 0.

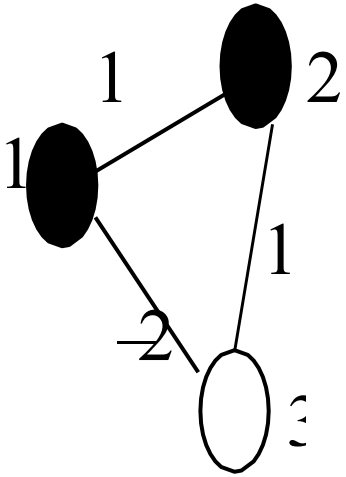
- *Repeat*
  - Choose any unit randomly. The chosen unit may be active or inactive.
  - For the chosen unit, compute the sum of the weights on the connection to the active neighbours only, if any.
    - If sum > 0 (threshold is assumed to be 0), then the chosen unit becomes active, otherwise it becomes inactive.
  - If chosen unit has no active neighbours then ignore it, and status remains same.
- *Until* the network reaches to a stable state

Current State	→ Selected Unit from current state	Corresponding New State
 Here, the sum of weights of active neighbours of a selected unit is calculated.		 Sum = 3 – 2 = 1 > 0; activated
		 Sum = –2 < 0; deactivated

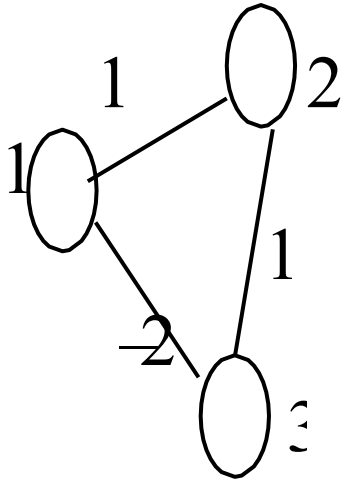
Stable Networks



X=[011]



X=[110]



X=[000]

Outline

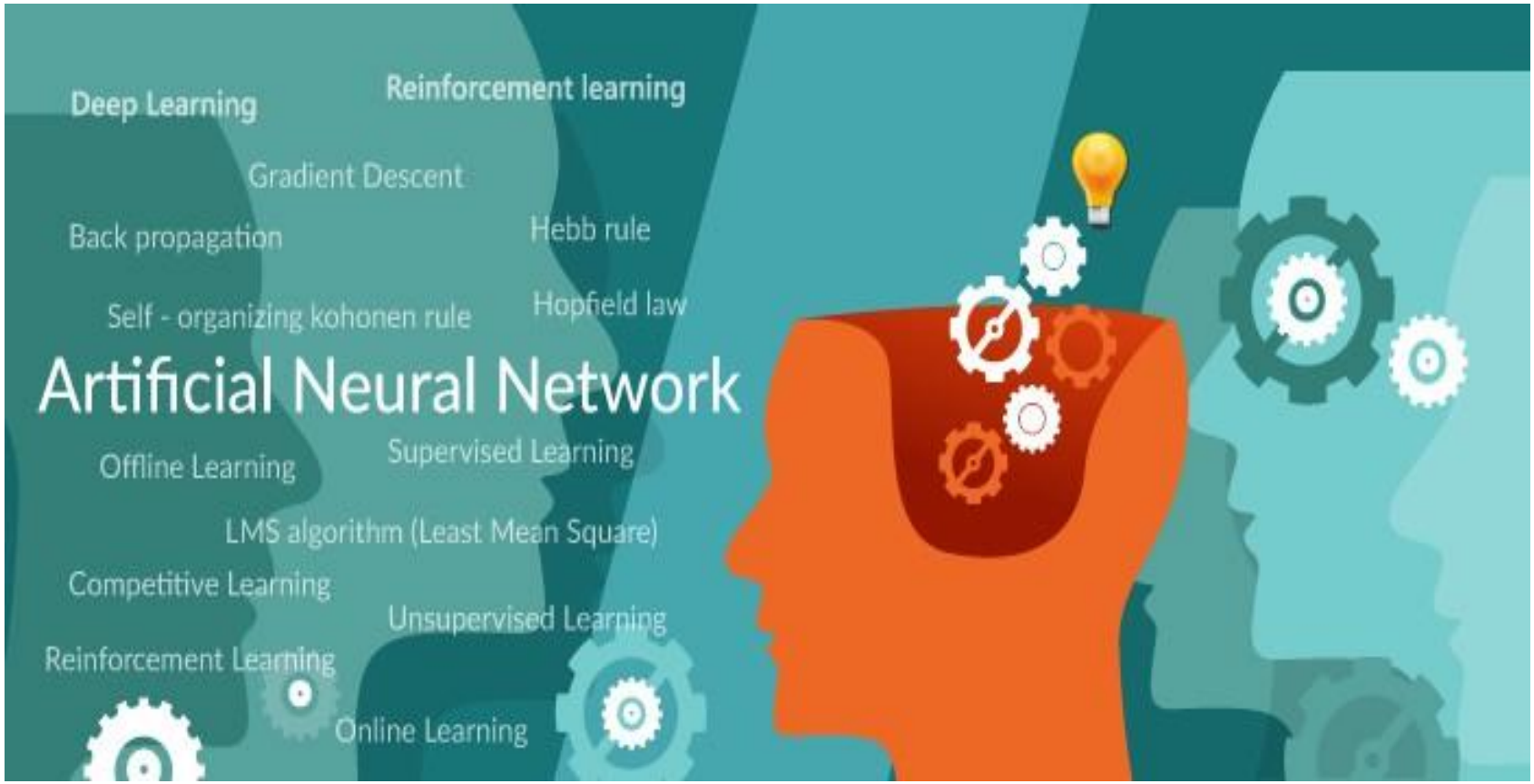
- What is Artificial Neural Network ?
- Biological Neuron
- Comparing ANN & BNN
- Model
- Architecture
- Learning
- Advantages & Disadvantages
- Applications
- Mathematical Representation
- Backpropagation & Stopping Criteria
- ANN Design Issues

www.fai.utm.my

Learning in Biology(Human)

- Learning = learning by adaptation
- The young animal learns that the green fruits are sour, while the yellowish/reddish ones are sweet. The learning happens by adapting the fruit picking behaviour.
- At the neural level the learning happens by changing of the synaptic strengths, eliminating some synapses, and building new ones.
- The objective of adapting the responses on the basis of the information received from the environment is to achieve a better state. E.g., the animal likes to eat many energy rich, juicy fruits that make its stomach full, and makes it feel happy.
- In other words, the objective of learning in biological organisms is to optimise the number of available resources, happiness, or in general to achieve a closer to optimal state

Learning in Artificial Neural Networks



Types of Learning

Supervised Learning

The training data is input to the network, and the desired output is known weights are adjusted until output yields desired value.

Unsupervised Learning

The input data is used to train the network whose output is known. The network classifies the input data and adjusts the weight by feature extraction in input data.

Reinforcement Learning

Here the value of the output is unknown, but the network provides the feedback whether the output is right or wrong. It is semi-supervised learning.

Offline Learning

The adjustment of the weight vector and threshold is done only after all the training set is presented to the network. It is also called batch learning.

Online Learning

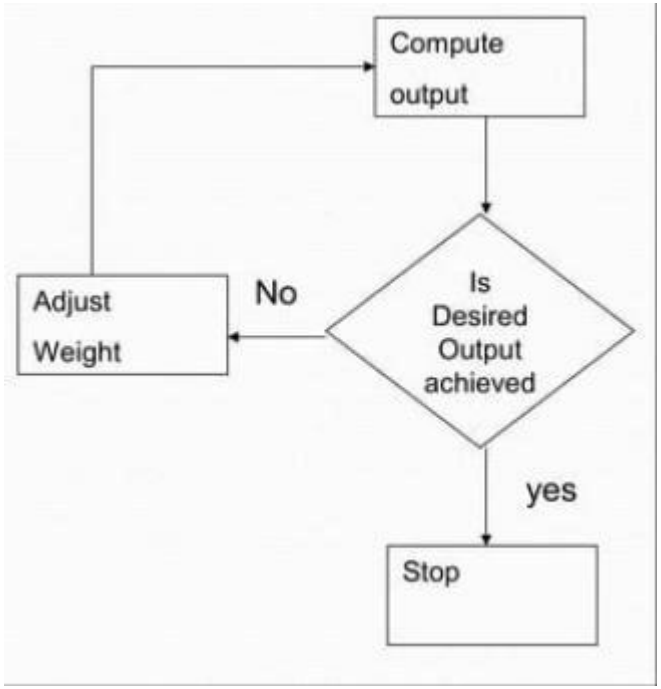
The adjustment of the weight and threshold is done after presenting each training sample to the network.

Learning Data sets in ANN

- **Training set:** A set of examples used for learning, that is to fit the parameters [i.e. weights] of the network. One Epoch comprises of one full training cycle on the training set.
- **Validation set:** A set of examples used to tune the parameters [i.e. architecture] of the network. For example to choose the number of hidden units in a neural network.
- **Test set:** A set of examples used only to assess the performance [generalization] of a fully specified network or to apply successfully in predicting output whose input is known.

Learning Process

- Learn from experience
  - -Learning algorithms
  - -Recognize pattern of activities
- Involves 3 tasks
  - -Compute outputs
  - -Compare outputs with desired targets
  - -Adjust the weights and repeat the process



Learning principle for artificial neural networks

- Learning occurs when the weights inside the network get updated after many iterations.
- For example—Suppose we have inputs in the form of patterns for two different class of patterns—I & 0 as shown and b -bias and y as desired output.
- We want to classify input patterns into either pattern ‘I’ & ‘O’.
- Following are the steps performed:
- 9 inputs from x1—x9 along with bias b (input having weight value 1) is fed to the network for the first pattern.
- Initially, weights are initialized to zero.
- Then weights are updated for each neuron using the formulae: $\Delta w_i = x_i y$ for $i = 1$ to 9 (Hebb’s Rule)

Learning principle for artificial neural networks

- Finally, new weights are found using the formulae:
- wi(new) = wi(old) + Δwi
- Wi(new) = [111–11–1 1111]
- The second pattern is input to the network. This time, weights are not initialized to zero. The initial weights used here are the final weights obtained after presenting the first pattern. By doing so, the network
- The steps from 1–4 are repeated for second inputs.
- The new weights are Wi(new) = [0 0 0 -2 -2 -2 000]
- So, these weights correspond to the learning ability of network to classify the input patterns successfully.

Outline

- What is Artificial Neural Network ?
- Biological Neuron
- Comparing ANN & BNN
- Model
- Architecture
- Learning
- Advantages & Disadvantages
- Applications
- Mathematical Representation
- Backpropagation & Stopping Criteria
- ANN Design Issues

www.fai.utm.my

Advantages Of ANN

- Parallel processing abilities mean the network can perform more than one job at a time.
- Information is stored on an entire network, not just a database.
- Fault tolerance means the corruption of one or more cells of the ANN will not stop the generation of output.
- Gradual corruption means the network will slowly degrade over time, instead of a problem destroying the network instantly.
- ANN can learn from events and make decisions based on the observations.

Disadvantages Of ANN

- There is no specific rule for determining the structure of artificial neural networks.
- The appropriate network structure is achieved through experience and trial and error.
- The display mechanism to be determined will directly influence the performance of the network.
- This is dependent on the user's ability

Outline

- What is Artificial Neural Network ?
- Biological Neuron
- Comparing ANN & BNN
- Model
- Architecture
- Learning
- Advantages & Disadvantages
- Applications
- Mathematical Representation
- Backpropagation & Stopping Criteria
- ANN Design Issues

www.fai.utm.my

Applications

- **Prediction: learning from past experience**
  - *pick the best stocks in the market*
  - *predict weather*
  - *identify people with cancer risk*
- **Classification**
  - *Image processing*
  - *Predict bankruptcy for credit card companies*
  - *Risk assessment*

Applications

- **Recognition**
  - *Pattern recognition: SNOOPE* (bomb detector in U.S. airports)
  - *Character recognition*
  - *Handwriting: processing checks*

- **Data association**
  - *Not only identify the characters that were scanned but identify when the scanner is not working properly*

Applications

- **Data Conceptualization**
  - *infer grouping relationships*
    e.g. extract from a database the names of those most likely to buy a particular product.

- **Data Filtering**

  e.g. take the noise out of a telephone signal, signal smoothing

- **Planning**
  - *Unknown environments*
  - *Sensor data is noisy*
  - *Fairly new approach to planning*

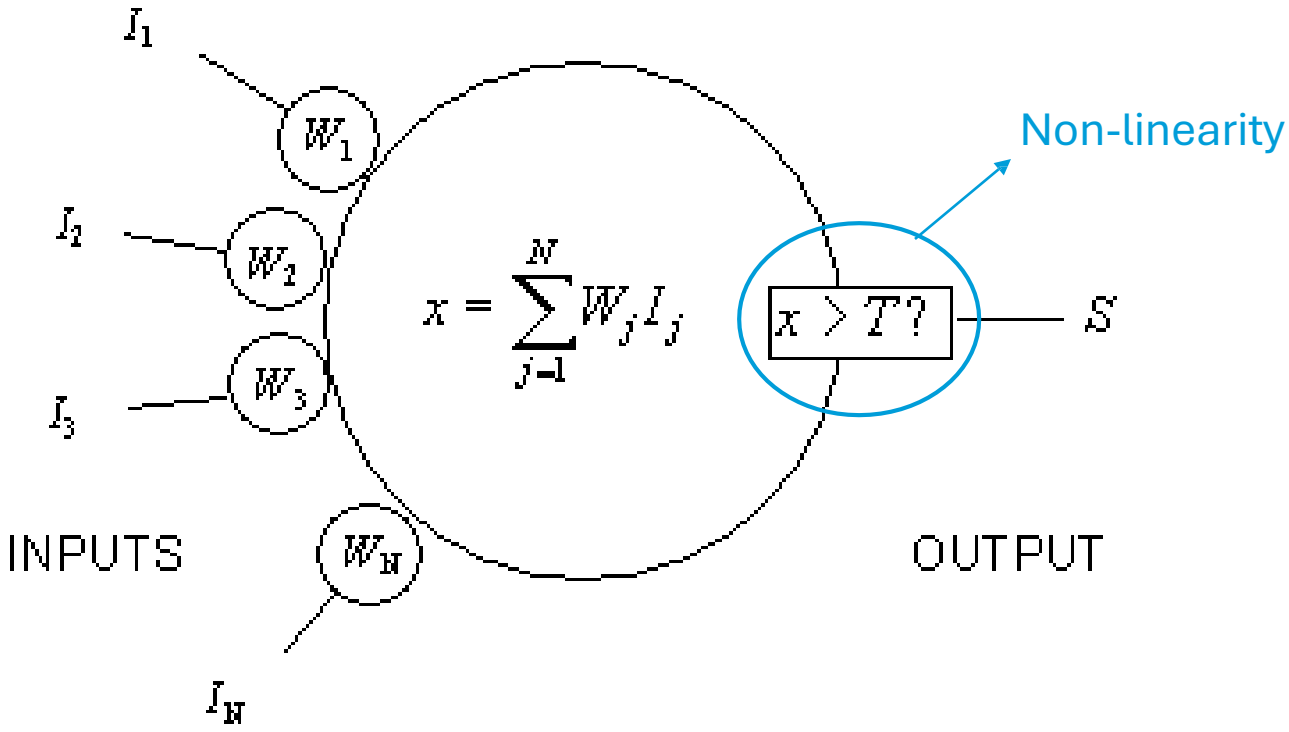
Outline

- What is Artificial Neural Network ?
- Biological Neuron
- Comparing ANN & BNN
- Model
- Architecture
- Learning
- Advantages & Disadvantages
- Applications
- Mathematical Representation
- Backpropagation & Stopping Criteria
- ANN Design Issues

www.fai.utm.my

Mathematical representation

The neuron calculates a weighted sum of inputs and compares it to a threshold. If the sum is higher than the threshold, the output is set to 1, otherwise to -1.



A simple perceptron

- It's a single-unit network
- *Change the weight by an amount proportional to the difference between the desired output and the actual output.*

$$\Delta W_i = \eta * (D-Y).I_i$$

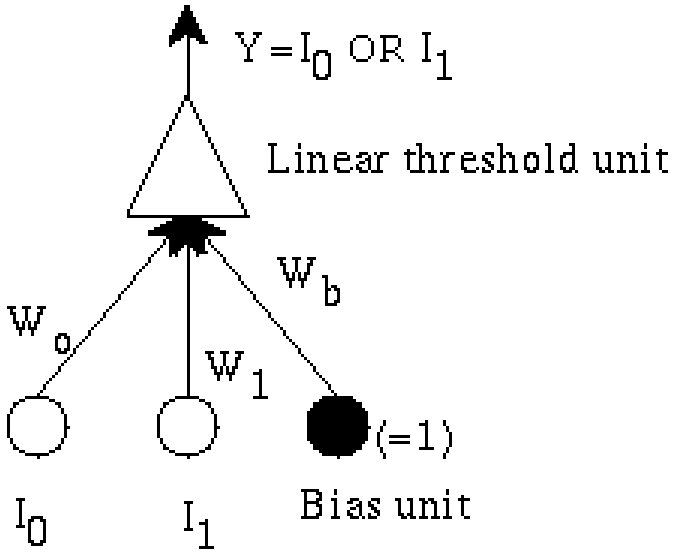
Input

Learning rate

Desired output

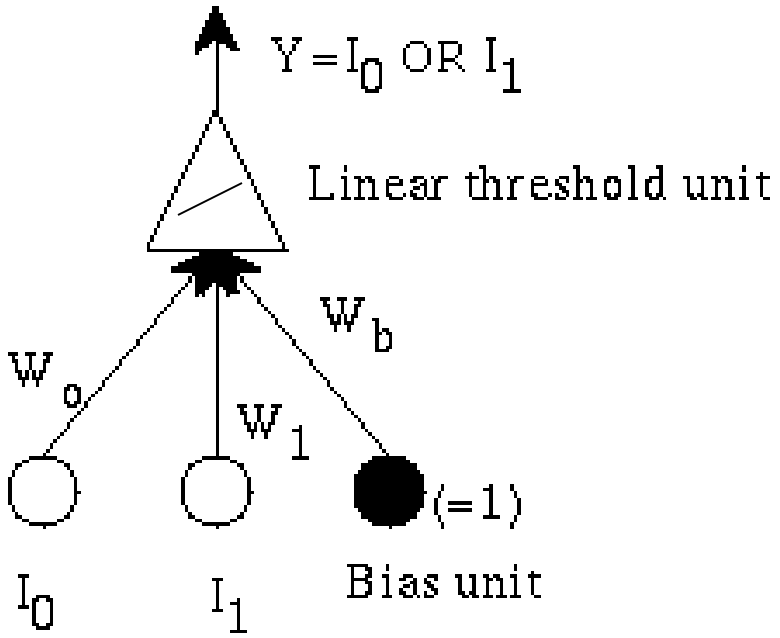
Actual output

Perceptron Learning Rule



Example: A simple single unit adaptive network

- The network has 2 inputs, and one output. All are binary. The output is
  - 1 if $W_0I_0 + W_1I_1 + W_b > 0$
  - 0 if $W_0I_0 + W_1I_1 + W_b \leq 0$
- We want it to learn simple OR: output a 1 if either I_o or I_1 is 1.



Outline

- What is Artificial Neural Network ?
- Biological Neuron
- Comparing ANN & BNN
- Model
- Architecture
- Learning
- Advantages & Disadvantages
- Applications
- Mathematical Representation
- Backpropagation & Stopping Criteria
- ANN Design Issues

www.fai.utm.my

Training Algorithm: Backpropagation

- It searches for weight values that minimize the total error of the network over the set of training examples (training set).

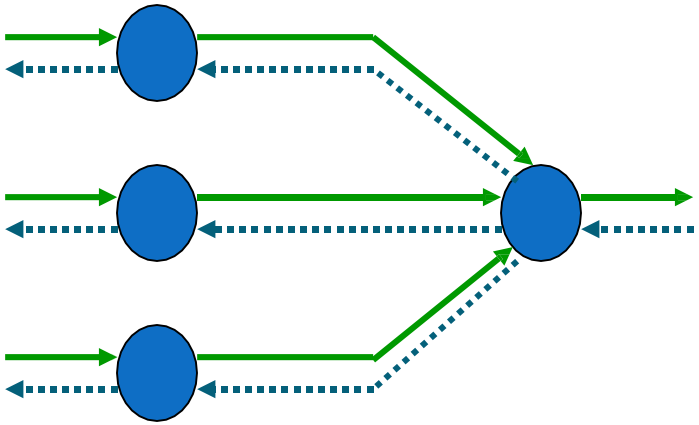
- Backpropagation consists of the repeated application of the following two passes:

  - **Forward pass**: In this step, the network is activated on one example and the error of (each neuron of) the output layer is computed.

  - **Backward pass**: in this step the network error is used for updating the weights. The error is propagated backwards from the output layer through the network layer by layer. This is done by recursively computing the local gradient of each neuron.

Backpropagation

- Back-propagation training algorithm



Network activation Forward Step

Error propagation Backward Step

- Backpropagation adjusts the weights of the NN in order to minimize the network total mean squared error.

Stopping criterions

- Total mean squared error change:
  - Back-prop is considered to have converged when the absolute rate of change in the average squared error per epoch is sufficiently small (in the range [0.1, 0.01]).

- Generalization based criterion:
  - After each epoch, the NN is tested for generalization.
  - If the generalization performance is adequate then stop.
  - If this stopping criterion is used then the part of the training set used for testing the network generalization will not used for updating the weights.

Outline

- What is Artificial Neural Network ?
- Biological Neuron
- Comparing ANN & BNN
- Model
- Architecture
- Learning
- Advantages & Disadvantages
- Applications
- Mathematical Representation
- Backpropagation & Stopping Criteria
- ANN Design Issues

www.fai.utm.my

ANN Design Issues

- Network Parameters
- Network Topology
- Choice of Learning Rate

Network Parameters

- How are the weights initialized?
- How is the learning rate chosen?
- How many hidden layers and how many neurons?
- How many examples in the training set?

Network Topology

- The number of layers and neurons depend on the specific task.
- In practice this issue is solved by trial and error.
- Two types of adaptive algorithms can be used:
  - start from a large network and successively remove some neurons and links until network performance degrades.
  - begin with a small network and introduce new neurons until performance is satisfactory.

Choice of learning rate

- The right value of learning rate depends on the application.

- Values between 0.1 and 0.9 have been used in many applications.